

Def. Let $A \in M_{m \times n}(F)$ be given where $F = \mathbb{R}$ or $\mathbb{C}$. Define the supremum norm of A

$$\|A\|_p \stackrel{\text{def}}{=} \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} \stackrel{(*)}{=} \sup_{\|x\|_p=1} \|Ax\|_p$$

where $\|\cdot\|_p$ is the p -norm, that is,

$$\|\cdot\|_p: F^n \rightarrow \mathbb{R}; (a_i)_{i=1}^n \mapsto \left(\sum_{i=1}^n |a_i|^p \right)^{\frac{1}{p}}.$$

The equality $(*)$ is given by the fact that for any non-zero vector x , then

$$\frac{\|Ax\|_p}{\|x\|_p} = \left\| A \cdot \frac{x}{\|x\|_p} \right\|_p = \|Ax'\|_p, \text{ where } \|x'\|_p = 1.$$

In fact, the value $\|A\|_p$ is the smallest value C such that

$$\|Ax\|_p \leq C \cdot \|x\|_p \text{ for all non-zero } x \in F^n$$

That is, given $A \in M_{m \times n}(F)$ and $x \in F^n$,

$$\|Ax\|_p \leq \|A\|_p \|x\|_p$$

Prop. Given $A \in M_{m \times n}(F)$ where $F = \mathbb{R}$ or $\mathbb{C}$,

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |A_{ij}| \quad \text{and} \quad \|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |A_{ij}|$$

Proof. Given $x \in \mathbb{R}^n$ with $\|x\|_1 = 1$, denote $x = (x_1, \dots, x_n)$.

$$\|Ax\|_1 = \underbrace{\left\| \sum_{j=1}^n x_j \cdot A e_j \right\|_1}_{\text{def}} \leq \underbrace{\sum_{j=1}^n |x_j| \cdot \|A e_j\|_1}_{\text{Hom + Sub. add}} \leq \sum_{j=1}^n |x_j| \cdot \max_{1 \leq i \leq n} \|A e_j\|_1 = \max_{1 \leq j \leq n} \|A e_j\|_1.$$

And, take $x = e_k$ such that e_k maximizes the $\|A e_j\|_1$, then we obtain the first.

Given $x \in \mathbb{R}^n$ with $\|x\|_\infty = 1$. Then, for any $1 \leq k \leq m$,

$$|(Ax)_k| = \left| \sum_{j=1}^n A_{kj} \cdot x_j \right| \leq \sum_{j=1}^n |x_j| \cdot |A_{kj}| \leq \sum_{j=1}^n |A_{kj}|$$

and the equality holds when $x = (1, 1, \dots, 1)$. So proved. $\square$

The $\|\cdot\|_2$ is so special, because it is induced by the inner product.

Note that the fact that for any unitary $Q \in M_{m \times n}(F)$, Q preserves the norm. That is,

$$\|Qx\|_2^2 = \langle Qx, Qx \rangle = \langle x, x \rangle = \|x\|_2^2$$

Thm. Given $A \in M_{m \times n}(F)$ where $F = \mathbb{R}$ or $\mathbb{C}$,

$$\|A\|_2 = \sigma_{\max}, \text{ the largest singular value of } A.$$

Proof. Given A , there is a Singular Value decomposition,

$$A = U \Sigma V^* \text{ where } U \in M_{m \times m}, V \in M_{n \times n} \text{ are unitary,}$$

$$\text{and } \Sigma = \begin{pmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{pmatrix} \in M_{m \times n}, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$$

so,

$$\|Ax\|_2 = \|U \Sigma V^* x\|_2 = \|\Sigma V^* x\|_2 = \|\Sigma y\|_2 \text{ where } y = V^* x$$

Since the unitary operation is isomorphism, and isometry, we can replace $V^* x$ into y .

And, the p -norm of diagonal matrix is the largest absolute value of its entry, so proved.

Problems.

For $A = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 2 \end{bmatrix}$, Compute $\|A\|_1, \|A\|_\infty, \|A\|_F, \|A\|_2$.

Proof.

$$\|A\|_1 = |-2| + |1| = 3, \quad \|A\|_\infty = |1| + |-2| = 3,$$

$$AA^* = \begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$$

$$\|A\|_F = \sqrt{1+4+1+4} = \sqrt{10} \quad (= \sqrt{\text{tr}(AA^*)}).$$

Meanwhile, $A^*A = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 2 \\ 0 & 2 & 4 \end{bmatrix}$ has a characteristic polynomial

$$\chi_{A^*A} = t(t-3)(t-7)$$

$$\text{So, } \|A\|_2 = \sqrt{7}.$$

Prove that Frobenius Cauchy-Schwarz inequality

$$|\text{tr}(A^T B)| \leq \|A\|_F \cdot \|B\|_F.$$

And solve $\max_{\|X\|_F \leq 1} \text{tr}(A^T X)$, find all maximizers for a non-zero A .